

BUFFALO 13 ESE, SD, USA

WMO: 726640

Lat: 45.5160N

Lon: 103.3017W

Elev: 2883

StdP: 13.23

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94081

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.1	-8.9	-21.2	1.9	-15.7	-14.7	2.8	-9.1	22.1	29.2	19.4	25.2	3.2	N/A	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.0	95.0	65.4	91.2	64.7	87.5	64.1	70.5	84.7	68.7	83.2	66.9	81.3	7.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	106.8	78.1	63.8	98.6	75.5	62.0	92.5	73.4	36.3	84.7	34.6	83.3	33.1	81.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.8	17.2	15.0	DB	-27.4	101.3	7.2	5.3	-32.6	105.1	-36.8	108.2	-40.9	111.1	-46.1	115.0	
			WB	-28.2	73.6	8.4	2.3	-34.2	75.2	-39.1	76.5	-43.8	77.8	-49.9	79.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	19.4	18.9	32.0	41.9	52.3	63.3	71.5	68.1	58.8	44.0	31.8	19.9
	DBStd	21.30	13.41	14.58	13.12	10.04	8.82	7.32	6.58	6.79	8.70	9.63	11.23	12.92
	HDD50	4478	949	871	566	275	78	3	0	0	20	232	550	934
	HDD65	8270	1414	1291	1023	692	403	114	16	47	222	652	997	1399
	CDD50	2155	0	0	8	33	148	402	666	562	286	47	3	0
	CDD65	473	0	0	0	0	8	64	217	144	38	2	0	0
	CDH74	7416	0	0	8	37	186	970	3054	2192	891	78	0	0
	CDH80	3241	0	0	0	7	39	333	1497	995	348	22	0	0
	WSAvg	6.2	6.3	6.0	6.9	7.6	7.2	6.3	5.6	5.4	5.7	6.1	5.5	5.7
	PrecAvg	15.0	0.4	0.4	0.8	1.4	2.3	3.0	2.2	1.3	1.1	1.3	0.5	0.4
(15) Precipitation	PrecMax	20.7	1.7	1.1	2.3	2.8	6.3	5.2	6.4	2.9	3.9	5.2	2.3	1.0
	PrecMin	8.0	0.1	0.0	0.4	0.3	0.7	1.6	0.3	0.2	0.1	0.2	0.0	0.0
	PrecStd	3.2	0.4	0.3	0.4	0.8	1.6	1.1	1.3	0.7	0.9	1.3	0.6	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.4	60.4	74.2	78.5	85.2	94.1	103.1	97.6	93.2	82.4	69.4	56.3
		MCWB	40.9	45.0	51.2	54.3	62.2	66.1	63.8	64.6	63.7	58.2	49.5	43.0
	2%	DB	45.9	50.3	66.0	71.7	79.0	88.0	96.6	93.7	88.3	74.2	60.8	47.1
		MCWB	37.6	40.2	46.8	52.7	60.0	65.0	66.5	63.8	61.3	54.3	45.5	37.6
	5%	DB	40.9	43.3	59.3	65.8	74.8	83.5	92.5	89.9	84.4	67.6	55.3	42.0
		MCWB	34.3	36.1	44.3	49.2	57.3	63.8	65.8	64.6	60.1	50.7	42.9	34.7
	10%	DB	37.6	38.3	52.3	60.8	70.2	79.6	88.4	85.9	78.9	62.1	49.7	37.6
		MCWB	33.0	33.4	41.5	46.7	56.0	62.5	65.3	63.8	58.6	48.6	39.9	32.7
	0.4%	WB	40.5	46.3	52.2	56.3	65.4	72.0	74.1	70.8	68.3	59.3	50.2	43.5
		MCDB	51.7	61.2	71.4	77.1	81.0	84.5	84.6	85.4	85.4	76.6	67.9	55.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.9	41.2	48.3	52.7	62.3	69.0	71.2	68.6	65.2	56.3	46.2	38.5
		MCDB	43.5	52.0	64.9	69.3	73.6	80.9	85.4	84.5	80.5	72.1	59.5	46.8
	5%	WB	34.6	37.1	45.4	49.8	59.7	66.4	69.5	67.1	62.7	52.4	42.6	35.1
		MCDB	39.4	43.2	60.3	63.5	71.4	79.2	83.8	82.4	77.5	65.0	53.2	41.8
	10%	WB	32.7	34.2	42.5	46.9	56.9	64.4	67.8	65.3	60.1	49.5	39.5	32.6
		MCDB	36.8	39.1	53.4	58.7	67.8	77.0	82.0	80.3	75.1	60.7	48.1	37.5
	5% DB	MDBR	21.9	22.4	24.0	24.9	24.1	25.5	29.0	29.9	30.3	25.9	25.2	21.9
		MCDBR	25.8	28.9	36.2	34.8	31.9	31.5	34.4	35.8	39.5	37.1	34.3	27.0
		MCWBR	16.9	18.5	20.0	19.0	16.2	14.4	11.2	13.5	15.7	18.9	21.1	18.0
		MCDBR	22.6	26.5	35.3	33.7	26.9	28.0	27.8	29.7	30.9	33.2	33.9	25.9
	5% WB	MCWBR	16.9	17.5	19.5	18.2	14.0	14.8	12.3	12.4	13.6	17.6	20.7	18.1
(40) Clear-Sky Solar Irradiance	taub		0.254	0.259	0.277	0.314	0.334	0.347	0.365	0.365	0.326	0.290	0.264	0.248
	taud		2.541	2.541	2.545	2.436	2.451	2.445	2.423	2.400	2.495	2.564	2.539	2.542
	Ebn at Noon		281	300	307	300	296	291	284	280	284	283	274	272
	Edh at Noon		20	24	28	34	35	35	35	35	29	23	20	18
(44) All-Sky Solar Radiation	RadAvg		569	889	1243	1591	1872	2149	2231	1884	1448	942	619	464
	RadStd		35	64	100	92	189	158	95	100	92	93	58	25

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page